Lecture 18: What Happens When Things Go Wrong: Mental Illness, Part I (full transcript)

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Overview: Professor Susan Nolen-Hoeksema describes how modern clinical psychology both identifies and treats various mental disorders. Particular focus is placed upon mood disorders such as bipolar disorder and depression, including current diagnostic criteria and current practices for treatment.

Chapter 1. Introduction to Dr Susan Nolen-Hoeksema [00:00:00]

Professor Paul Bloom: I am extremely pleased to introduce the fourth and final guest lecture of the semester. Professor Susan Nolen-Hoeksema. Susan is a professor in the Department of Psychology and the Director of Graduate Studies. She is well known for her work in clinical psychology and especially her research in depression, the nature and causes of people with depression, with special focus on sex differences in depression. She basically does everything someone can do. She is a noted scientist, winning many awards and publishing massive amounts of work in scientific journals. She is an award-winning teacher and has authored what, in my mind, is the very best textbook in her area. And she's a noted popular writer who has written popular and accessible books bringing the message and ideas and theories of clinical psychology to the broader public. The only other thing I'll mention before we welcome her is that she's going to teach next year her course in clinical psychology, which has a superb reputation as an extremely interesting course. If you are interested in what you hear today and you want to learn more about it, that's the course you should take. So, let's please welcome Dr. Susan Nolen-Hoeksema. [applause]

Chapter 2. Behavioral Criteria for Accessing Mental Disorders [00:01:36]

Professor Susan Nolen-Hoeksema: Thank you Paul. Can everybody hear me okay? Okay. So, what I want to do today is to give you a very brief overview of how modern clinical psychology looks at mental disorders, some of the ways we think about what constitutes a mental disorder, some of the characteristics that kind of cut across mental disorders, and then I'm going to use the case of mood disorders, that is depression and what is now called bipolar disorder, what you may know more popularly as manic-depression, as sort of examples of how we think about a particular set of disorders and some of the ways we go about researching the theories — different theories for the disorders and some of the prominent treatments for disorders these days. Okay? So, I'm going to do both a fair amount of lecturing, and then I've got lots of video clips to show you as well. So, I'm going to be roaming around and changing venues here fairly often.

So, the first and most fundamental question in clinical psychology is, "What is abnormality?" Where do we draw the line between normal, healthy, typical behavior and what we might want to call abnormal, atypical, deviant, unhealthy, maladaptive mental problems? We tend to have an intuitive sense of what we mean by abnormality, and we'd like to believe — a lot of people who come into my course say, "Well, of course, you know, you guys have figured it out. You know where to draw the line. You have

criteria. You have blood tests, right? — that tell me whether I have depression or schizophrenia or one of the things I've read about." Well, the reality is that we don't.

First of all, there is no biological test for any of the known mental disorders right now. And instead what we have is a set of behavioral criteria for how to diagnose different mental disorders. And what I mean by behavioral criteria is a set of symptoms that the person reports to you about how they feel, how they think, and a set of observations about their behavior and how typical or atypical it is. And you take the sort of set of symptoms the person shows or reports, and you match them up against the existing criteria for different mental disorders. And then it comes down to a fairly subjective judgment call about whether the person meets the criteria or not. Unfortunately, these judgment calls, because they are so subjective, can be influenced by a lot of factors. And we won't have a chance to go into these too much today, but just to highlight a few of them.

The first is social norms. Whether you get labeled as having a mental disorder or a problem depends very heavily on what your social or cultural norms are. So, a woman wearing a veil in a Muslim community or culture would be seen as typical, even prescribed, behavior. Whereas a woman wearing a veil in a non-Muslim culture, especially until fairly recently, was often looked upon as very atypical or abnormal behavior.

The second kind of thing that gets — that influences whether something is called normal or abnormal is certain characteristics of the target person. In particular, I've highlighted here, gender. Whether you're a man or you're a woman really influences how unusual a certain behavior is. So, crying is a good example. A man crying in our culture is seen as fairly unusual, whereas a woman crying is seen as much less unusual. On the other hand, a woman beating up someone is taken as quite unusual behavior where it's less unusual for a man. So, we have gender stereotypes, gender roles for what is acceptable behavior, and our judgments as to whether something is normal or abnormal get influenced by those gender roles.

And the third thing that can influence whether something is labeled abnormal or not is the context. And here I'm giving you the example of "paranoia." If you're paranoid and hyper-vigilant, looking for threat in downtown Baghdad, that's considered very adaptive behavior these days because it could prevent you from getting hurt or killed. Whereas, if you're in a quiet little farm in Central Connecticut, being extremely paranoid and believing there's someone who's going to shoot you around the corner is not considered as normal or as acceptable or adaptive behavior. So, the context in which you exhibit a particular behavior also can heavily influence whether it gets labeled by others as normal or abnormal.

In the field of clinical psychology we have a number of different ways, kind of heuristics that we use to label things as abnormal or unhealthy or troubling. And three of these characteristics are what we often call the three Ds: distress, dysfunction, and deviance. So, behaviors that cause the individual or others significant distress often get labeled as abnormal or unhealthy. Depression is a prime example, as we'll see when we talk about the characteristics of it. It's a miserable state of being; you're unhappy, you're sad, you may even feel so badly you want to kill yourself. And that very, very high level of distress is part of the reason why it's labeled as a mental disorder. Other mental disorders don't cause the individual distress, but they may cause other people distress.

So, one example of this is something called "antisocial personality disorder," where the individual has no regard for the rights of other people, has no hesitation to steal or — steal from or hurt other people, has no empathy or sympathy for other people's feelings and so can inflict a lot of harm on other people and has absolutely no distress over this whatsoever. But this behavior causes other people distress, and that's one of the reasons why that's labeled an abnormal behavior or a mental health problem.

The second general criterion is "dysfunction." If a set of behaviors prevents the person from functioning in daily life, then it might be labeled as abnormal or might end up being labeled as a mental health problem. Again, depression is a good example. People who are depressed often become completely non-functional. They can't get up and go to class; they can't go to work; they can't interact with their friends; they withdraw and become totally isolated socially. So, they might lose their job; they might flunk out of school. And this complete decline in functioning is one of the major reasons that we consider depression one of the most debilitating disorders.

And then finally, "deviance," the behaviors or feelings are highly unusual. This is probably the most controversial of the three because it weighs, it is so heavily influenced by the social norms. What's deviant in one culture is not deviant in another culture. But if a set of behaviors is completely unacceptable to a culture, highly unusual, they're more likely to end up getting labeled as abnormal.

Okay. So, how do we pull this all together? Well, these days the manual for making diagnoses in clinical psychology and psychiatry in the United States is called the Diagnostic and Statistical Manual or the DSM, and it's in its fourth revision. It's been around since the, I believe the '50s, and the early editions in the '50s and '60s were highly subjective and based on Freudian theory. But since 1980 there's been real effort to make the criteria much more objective, to make the set of behaviors or observations that are required to diagnose someone be things that are observable, that you can see in other people that they can report on reliably, and that one clinician and another clinician will agree upon. So, the DSMgives lists of symptoms with the required symptoms for a diagnosis, the number of symptoms that have to be present, and the notions of deviation, dysfunction and distress are built into these criteria. And I'm going to give you a couple of examples of these criteria when we talk about the specific types of mood disorder.

So as I said, I'm going to use mood disorders as kind of a case example here of how we go about diagnosing and understanding psychopathology, but I also just want to impart some information because mood disorders are one of the most common problems that people face. As many as one in four women will have an episode of serious depression at some time in her life, and about 13% of men will have an episode of serious depression in their lives. So, these are extremely common kinds of problems that people experience, particularly at your age. The college years are one of the peak times of onset, first onset, of depression in particular. And also, for bipolar disorder, or manic-depression, the late adolescent, early 20s are the peak onset times for these disorders as well.

Chapter 3. Unipolar Disorders [00:11:54]

So, the mood disorders divide into what's called unipolar depression disorders, which is depression only and then bipolar disorders where the person cycles between depression and mania. And here are the DSM criteria for major depression, one of the most severe forms of depression. And as I said, the DSM sets up these relatively observable criteria and how many you have to have and what absolutely

has to be present in order to get the diagnosis. So, the first criterion in the DSM for major depression is that the individual has to either show sadness or a diminished interest or pleasure in their usual activities, which is referred to as anhedonia. So, you have to have one or the other of these to sort of pass the first criterion. So, you might say that you feel sad and blue and just — or actually say you feel depressed. Some people feel those feelings very strongly. Other people don't really feel so sad or blue, but what they'll say is that nothing interests them anymore. It's like the emotion has been sucked out of their life altogether. They don't have any fun doing the activities they used to do before. They don't want to hang with their friends. They just — they don't care about eating. Just nothing feels right, feels good, anymore.

And then the individual has to have four of the — at least four of the following symptoms in addition to sadness or anhedonia. First, they can show significant weight or appetite change. So, you may completely lose your interest in eating and lose a lot of weight, or some people go on eating binges. I had a very good friend who was depressed for about a year, and she gained fifty pounds because she would just eat. She would binge eat, especially at night.

There are sleep disturbances — insomnia, which is having trouble sleeping, or hypersomnia, which is sleeping all the time. There's a particular form of insomnia that's especially likely in depression where you can go to sleep at night, but then you wake up at about three or four in the morning every night and you can't go back to sleep at all. You're just up for the rest of the night. But other people want to sleep all day long, and in the clip I'm going to show you in just a minute the woman talks about sleeping twenty, twenty-two hours a day, getting up, eating a little bit, and then going back to bed because she was exhausted still.

The third criterion is psychomotor retardation or agitation. The retardation is much more common, and what this means is that sort of everything about the person's movement is slowed down. They'll walk more slowly. Their reaction times will be slowed down. And because they're so much more slow moving, depressed people are often more prone to accidents. They just can't react as quickly as they need to when they're driving or when they're crossing the road and a car is coming at them suddenly. So, they get into more accidents. And their speech may be slowed down. They may talk very, very slowly and it's as though it just takes a tremendous amount of energy to get even a common sentence out. A much more, much smaller number of people get agitated instead of slow down. They may be hyper and just feel like they can't sit still and such, but the agitation is much more rare than the retardation. People feel really tired, fatigued and like they have absolutely no energy. They can't get up and can't get moving. As I said, they may want to just sleep all of the time.

Number five is feelings of worthlessness or excessive guilt. They may feel as though everything is their fault, and the guilt feelings or sense of worthlessness can even get psychotic. They can lose touch with reality. When a person loses touch with reality when they're depressed, it typically has really depressing themes. They may believe that they are Satan and that they have to commit suicide because they're inflicting evil on the world. They may believe as though random events are their fault, you know, that a flood that just happened somehow they caused. So, the feelings of worthlessness and guilt can get completely out of touch with reality, psychotic. More commonly, they're just unrealistic. They're negative self-esteem, just being down on yourself, feeling stupid and worthless and ugly and bad.

Number six is diminished ability to concentrate or indecisiveness. When you are depressed it's really hard to pay attention. You'll read a passage over and over again and you just can't process it at all. You can't concentrate on a lecture so going to class is just useless. You have to make a decision about what a paper topic is, and it just seems like the most monumental thing on earth. You just can't decide anything, you can't think anything; your thoughts are completely clouded and overwhelmed.

And then suicidal ideation or behavior; it means you think about committing suicide, you think about dying. And a subset of people actually take action to try to hurt themselves or kill themselves. Now, it should be said that suicidal thoughts and behavior don't only happen in depression. They actually happen in all types of psychopathology, but they're particularly common in depression.

So, you have to have at least one — four of those symptoms plus sadness or anhedonia, and these symptoms — it can't just be a bad day that you're having. These symptoms have to be present persistently for at least two weeks to get the diagnosis. Now, truth be told, most episodes of major depression actually last a lot longer than two weeks. In fact, the average length of an episode, if it's not treated, is at least six months. So, people stay this miserable for a very long period of time, but the minimum criterion in the DSM is at least two weeks.

So, what I want to do is to just show you a short clip of a woman who has had a lot of episodes of depression. Fortunately, at the moment she's not in an episode. But she can speak very articulately about what it's like to be in the midst of an episode and some of the significant symptoms that she had. [video clip plays]

Okay. There are couple of things she talks about that I just want to comment on. One is this differentiation between everyday sad mood and the kind of depressions we all experience and the kind of debilitating, overwhelming depression that she experiences. And it is true that there is this continuum from getting bummed out because you didn't do well on a test or because you broke up with a boyfriend or girlfriend or something like this and being completely not functional, vegetative, the way that this woman becomes whenever she gets depressed. And it would be nice if we were really sure where the cutoff was between those normal everyday depressions and what's really a disorder. But the reality is we don't really have real clear demarcation lines. There are a lot of people who have more moderate forms of depression than Tara here talks about but who still would qualify for a diagnosis and are still suffering and impaired by their symptoms. So, I don't want you to get the sense that if you don't have the kind of horrible version on the extreme end of the continuum of depression that Tara has, then there's nothing wrong with you, because that's not the case. People who are really slowed, whom their functioning is interfered with — they're just really unhappy with life — have problems that can be helped and do need attention. And it is the case that much more moderate forms of depression can morph into more serious forms if they're left untreated. So, there is this continuum.

The other things I wanted to comment on that she talks about early on in this piece is the fact that she hauls herself up and goes through her day, even when she's feeling really, really depressed. And there is this characteristic of a lot of depressed people that I call the "walking wounded." They just haul themselves through the day trying to act normal, trying not to let anybody know that there's anything wrong with them, trying to keep up with their schoolwork or their employment. But they're miserable and they're not functioning at the level that they're capable of and such. And that's something that's very, very common, and it's in part because people don't feel as though they should have to get

treatment or they're ashamed of getting treatment or seeking help for depression. And so, they just keep going on and going on, sometimes for years, in a very sorry state before — sometimes they just — they end up actually falling apart to the point where they have to get help.

Chapter 4. Bipolar Disorders [00:21:30]

Okay. The other category of mood disorders that I mentioned is bipolar disorders. And as I said, bipolar disorder involves symptoms or periods of depression but then also distinct periods of the opposite of depression, which we call "mania." So, the person cycles back and forth between debilitating depressions and manic episodes. So, let me describe manic episodes to you now.

So, the first criterion is that instead of feeling down, blue or depressed the person has an abnormally and persistently elevated expansive or irritable mood that isn't just, again, a good day because you won a prize or got an "A," but rather, it's this unusually positive, expansive mood for at least one week persistently. And then the person has to have three or more of the following symptoms.

First, inflated self-esteem or grandiosity. The individual may feel as though they are the smartest, the most creative, insightful, powerful person on earth, and they are perfectly happy to tell you this. So, there is no problem with self-esteem, thank you very much. "If you can't keep up with me it's your fault." There's a decreased need for sleep; they may only sleep a couple of hours a night and get up raring to go. They tend to be more talkative than usual, and there's a really pressure to their talk. They'll talk really pressured, and they'll talk really, really fast. And one of the reasons they're talking really, really fast is they have this flight of ideas. The thoughts are just racing through their mind, and they can't talk fast enough to get them out. And if you can't follow them, that — well, that's because you're not smart enough to follow them. But they've just got too many good ideas and they've got to get them out.

They're highly distractible. And then there is this increase in this — what the DSM calls this "goal-directed activity." Out of their grandiosity will come these grand schemes for — often for making a lot of money and they'll pursue these with great vigor no matter how totally irrational they are. So, it's not at all uncommon for them to cash out all the family bank accounts, to sell the house, to sell the car, to sell the kids so that they can finance this great scheme for making a zillion dollars on the Internet tomorrow. Right? Okay. And they'll pursue this with tremendous vigor.

They'll also get involved in all kinds of, what the DSM discreetly calls "pleasurable but dangerous activities." There's a lot of sexual promiscuity, a lot of drug abuse, a lot of, as I said, getting — going and gambling, believing that you're on a hot streak, there's nothing can stop you. You know, you're just so brilliant and you've got this scheme, you've got the plan. You're going to make it. Okay? So, the individual has three of more of these kinds of symptoms plus this elevated, expansive and often quite irritable mood. It's not just that they're happy, you know, and sort of upbeat. It's just that they're just impatient and irritable and trigger-fire. And sometimes they can become violent because they're just — they're so incredibly agitated and irritable.

So, let me show you a couple of clips. I have one really short one. It's not a real high quality clip, but it's a very, very nice example of an older woman who is in the midst of a manic episode. And it shows what

this kind of agitation and flight of ideas and racing thoughts can look like. And then the other one is — I'll introduce whenever you — we do it. [video clip plays]

Okay. So while she was at the hairdresser's it's a nice example of how she was just pressured to speak. Nobody was telling her that she had to say all of these things. They were just standing there with a camera and she was going on and on and on about these things. And as you could see she was getting more and more agitated and more and more irritated as she was retelling her story. And then what? In the last little bit there you saw her flip from her mania into a more depressed state. And this poor lady, unfortunately, is having a hard time finding a stable point. He talked about lithium there, and I'll talk about it in a little bit — about the use of lithium as a drug to try to stabilize these mood swings. But at this point in this video, it's not working for this lady. And so, she's flipping back and forth, but they're having a hard time finding that middle ground.

So, I want to show you another clip — it's a little bit longer — of a man who has bipolar disorder. He is not currently in an episode of either depression or mania, but again, he can talk about some of the things he got himself into and how it manifested in his behavior. [video clip plays] Hypomania is a more mild version of mania, but it's the same symptoms. [video clip resumes]

Okay. Just a couple of things that Bernie talks about that I want to comment on. One is that just as in depression, mania has — runs along a continuum. So, it can be relatively mild all the way to extremely severe and even psychotic. So, when a person with mania loses touch with reality, instead of having beliefs that they are Satan or they've done some horrible thing, they'll believe that they are some supernatural being. They may believe that they are the Messiah or that they are Albert Einstein, you know, come back to life, or that they have supernatural powers or something of this sort, so that their false beliefs, their delusions and their hallucinations, the things that they see and hear that aren't really there tend to be very grandiose in their themes. Bernie's mania is not on the far end of the continuum by any stretch, but you can see it still gets him into trouble.

Now, there are people who have — who cycle between fairly low levels of mania and fairly low levels of depression, back and forth. And there's been some argument that people who are kind of chronically, mildly manic — especially if they're really smart or they have a special talent – can make it work for them. And there is a wonderful book by Kay Jamison, who is a professor at Johns Hopkins where she chronicles — She does sort of historical biographies on a number of well-known authors and poets and musicians, Robert Schumann and a number of politicians, Winston Churchill and such, arguing that they actually had mild forms of bipolar disorder and that they were able to sort of channel the manic episodes through extraordinary talent or intelligence in ways that made it work for them. There are also a number of arguments that very, very successful CEOs sometimes are people who are chronically slightly manic. They can go on a couple of hours a night of sleep; they're obviously really quite grandiose and self-confident, and that they can maintain this kind of moderate level of mania, keep it under control and channel it in ways that work for them. So, if you're interested in that book, send me an email and I'm happy to send you the citation for it.

But for the most part, mania can get people into tremendous trouble. They can, as I said, get involved in sexual promiscuity that puts them at risk for sexually transmitted diseases. They can get involved in drug activity. They can get themselves arrested. They can certainly send themselves and their family into bankruptcy. And these kinds of negative consequences of the mania often are what motivates the

person to get help because the mania itself can be rather pleasurable to have. Also, what motivates them to get help is the plunge into depression, the knowledge that they will, at some point, come out of the mania and go into a debilitating depression.

Bipolar disorder is much less common than depression. I said that about 22% of women and about 13% of men will have an episode of serious depression at some time in their lives. Bipolar disorder occurs in only about 1% of the population, and it's equally prevalent in women and men. So, it's a really quite different disorder in many ways from depression alone.

Chapter 5. Statistics for Depression [00:30:20]

I want to give you some other statistics about depression per se, and this only applies to depression. There are quite large age differences in the prevalence of depression. These are data from a nationwide study of people between the ages of fifteen and fifty-five, and these are the percentages of people in this study. And there were several thousand people in the study. These are not people who have sought treatment for depression but just a random community sample. And this is the percentage who've had an episode of major depression in the past month. And as you can see, the fifteen to twenty-four age range has the highest rates, and then they go down somewhat, although the thirty-five to forty-four is fairly high as well with age. You might be surprised to learn that the rates of major depression in the elderly are actually quite low by most national statistics. And that's true up to about age eighty or eighty-five. And the arguments for why this is the case are very interesting.

There are some people who argue that as you get older you get wiser, and so that's why we see lower rates of depression in older age. There are other people who argue that current generation — younger generations now; your generation and the one above you — are more prone to depression and will be for the rest of your life compared to your grandparents, because of historical changes in the kinds of social support and family networks available and a number of other historical cultural changes. The other sort of side of the argument is that because depression is known to impact negatively your physical health – depression is associated with higher rates of cardio-vascular disease, stroke, immune system diseases, a whole host of diseases that people die from — that people who have a lifelong history of depression are actually more likely to die at an earlier age, and that's why we see relatively low rates in older age people. We don't know yet which of these explanations is true. It may be that they're all true to some extent.

There are also gender differences in depression. These are data from a compilation of hundreds of studies of children and adolescents, looking at not full-blown depression but levels of depression on self-report questionnaires. Probably most of you have filled out these questionnaires, like the Beck Depression Inventory that ask you how you've been feeling in the last month. And there's a kiddy version of this, and these are data from that, from several thousand children. And as you can see here, prior to the age of about thirteen, boys and girls have relatively similar levels of depression. But beginning around the pubertal years, girls' rates of depression go up quite dramatically and boys' rates stay the same or go down. And by the time they're eighteen or twenty you get almost a two-to-one ratio of depressed girls to depressed boys. And then this is true for the rest of the adult age span.

There are lots of hypotheses about this, why it's true. There are biological hypotheses that have to do with hormones. There are sociological hypotheses that have to do with the kinds of stress, and

particularly abuse, in girls' lives relative to boys'. We don't know exactly why. It's probably a lot of these things coming together that make this huge two-to-one ratio true.

Chapter 6. Biological, Cognitive and Interpersonal Theories and Treatments [00:34:04]

So, let's talk a little bit about the major theories and treatments for the mood disorders. There are biological theories and treatments, what are known as cognitive behavioral theories and treatments and then interpersonal theories and treatments. And I'll walk you through examples of each of these.

So, first, genetics. It's pretty clear that genetics are involved in the mood disorders, especially bipolar disorder. There is very strong evidence in bipolar disorder, and there are a number of ways — Have you talked about how you do genetic studies? Okay. So, you guys know about twin studies for example and family history studies. So, this is actually a compilation of a group of studies. And here you've got some of them – twin studies — compared. So, in monozygotic twins, if your identical twin has bipolar disorder, you have over a 60% chance of having the disorder yourself. In contrast, if it's your — if you're just a fraternal twin of a person with bipolar disorder, you only have about a 12% chance of having the disorder. So, that massive difference there is very strong evidence that there's a genetic component to the transmission of the disorder. Similarly, the more distant you get in terms of your biological relation to a person with bipolar disorder, the lower your rate or your risk of the disorder is. So, the second degree relatives of a person with bipolar disorder only have about 2% chance of getting the disorder. And that's barely above what's in the general population, which is about a 1% chance of getting the disorder. So, it's very clear that bipolar disorder has a genetic component to it.

With depression alone, major depression, there are probably versions of the disorder that have a stronger genetic component to them than others. And in particular, folks who have what's called "early onset depression," where their first episodes come on in childhood or very early adolescence, seem to have a form of depression that has a stronger genetic component to it. Whereas, people who have depression that is clearly triggered by a major life event like a trauma or a loss — those types of depression are less clearly linked strongly to genetic factors.

There are also a number of neurotransmitters that have been implicated in the mood disorders. And the class of neurotransmitters that's been researched most often is what's called the monoamines. I'm sure you've heard about the link between serotonin and depression, but there are two other monoamines, norepinephrine and dopamine, that have also been linked to both of the mood disorders, both bipolar disorder and depression. And it used to be thought that it was just that in people with depression they didn't have enough of these neurotransmitters, enough serotonin, in particular, in the system, in the brain, in order to function normally. But now the theories on what the role of neurotransmitters is have a lot more to do with the receptors for these neurotransmitters and their functioning. And the notion is that the receptors for neurotransmitters like serotonin don't function efficiently. So, even if there's enough of the chemicals in the synapses in the brain, the neurons can't make use of them because the receptors aren't functioning appropriately. And so what the drugs that help relieve depression do is to improve the functioning of these neurotransmitters.

There's a very interesting line of work that's going on right now looking at the intersection of genetic predisposition neurotransmitter functioning and stress. And we have one of the world's experts on this

kind of work now here at Yale, Julia Kim-Cohen, who just joined us in the last year. But there are several recent studies. There's another person in psychiatry, Joan Kaufman, who's done some of this work. But there are several recent studies that find that certain variations or polymorphisms on the serotonin transporter gene predict who will become depressed in the face of stress.

So, a classic study was done by Avshalom Caspi and colleagues, and they found that people who have one or two of what's called the short allele on the serotonin transporter gene — so this is a particular variation on the serotonin transporter gene — if they had one or two of these short alleles and they were confronted with stress, they were more likely to develop depression. But it's important to sort of dissect this. So people who had — it didn't really matter which of these genes you had. If you were never confronted with major kinds of stress, like maltreatment, if you weren't confronted with stress, you were no more likely to have depression regardless of what kind of serotonin gene you had. But if you had either one short allele or two short alleles and you were confronted with maltreatment as a child, you had a much greater probability of becoming depressed at some time in your life. And this has been replicated with other samples, with other forms of major trauma. And basically the story is, it takes the intersection of a genetic predisposition and major stress to create full-blown depression in some people. Now, that may not be true for all genetic predispositions or all forms of depression, but this serotonin finding has actually been replicated now in at least four different studies. So, it seems to be a pretty reliable effect. So again, genes do not determine the disorder, but the intersection of genes and stress seems to be a major risk factor for the disorder.

There are a number of brain areas that seem to be involved in the mood disorders where there is just dysregulation or dysfunction. The prefrontal cortex, as you probably have studied, is an area of the brain that's very involved in higher order complex thinking and problem solving and in goal-directed behavior. In people with depression, there's lowered activity in the prefrontal cortex, suggesting that — which may play a role in the difficulties in concentration, in goal-directed behavior, in planning and problem solving and in regulating emotion.

The amygdala is an area of the brain that is involved in the processing of emotional information. And people with mood disorders show overactive amygdala responses to emotional information. This is true in both bipolar and in depression. The hippocampus is an area of the brain that's very involved in memory and in concentration. And in people with chronic depression, you often see shrinkage in the hippocampus, and this may be related to their problems in concentration and attention. And then finally, the anterior cingulate is an area of the brain that's involved in a lot of different activities, but pertinent to the mood disorders, particularly in responses to distress — to stress and in sort of the choice of behaviors. And it may be that dysregulation of the anterior cingulate may be involved in the person's difficulty in responding appropriately to stress, in choosing good coping behaviors and changing their behaviors whenever their behaviors aren't working well.

So, from the biological theories come a number of different drugs to treat the mood disorders. Two of the older classes are called the monoamine oxidase inhibitors and the tricyclic antidepressants. The tricyclics are still used these days to some extent. They're relatively effective. About 60% of people respond well to the tricyclics, but they have a lot of side effects, and they can be fatal in overdose. And so there was a — has been always a search for other alternatives to them. The drugs that have really taken over the market are the selective serotonin re-uptake inhibitors or SSRIs. This is Paxil, Prozac

and the like. They were introduced in 1987 — Prozac was — in the U.S. market and truly took over the market in the treatment of depression and anxiety and a number of other disorders. Now, they're not that much more effective than the sort of old style antidepressants, but they have fewer side effects and they tend to be easier for people to tolerate.

More recently, there are selective serotonin/norepinephrine re-uptake inhibitors. These drugs, by the way, what these drugs supposedly do is to prevent the re-uptake of serotonin or serotonin and norepinephrine back into the sending neuron. So, it creates more of the neurotransmitter there in the synapse. And these are reasonably effective drugs, although, again, there's a substantial number of people who don't respond to them, especially immediately. And they'll often have to cycle through a number of drugs before they find one that works.

Lithium is the drug of choice for the treatment of bipolar disorder. It seems to stabilize the mood swings by stabilizing the number of different neurotransmitter systems. But the lithium is problematic because there are tremendous side effects. It's also dangerous for women to take while they're pregnant in terms of fetal development. So, it's a very tough drug to stay on. There are lots of gastrointestinal side effects and such, and people are often on lithium and the antidepressants because lithium often only affects the manic episodes but it doesn't really relieve the depression.

And then finally anti-psychotic medications, that is those help people who've lost touch with reality, are sometimes used to treat the mood disorders whenever the person has lost touch with reality.

I'm going to go through and talk about some of the psychosocial treatments because I want to get to them as well. The cognitive behavioral therapies are based on Aaron Beck's Negative Cognitive Theory of Depression. Aaron Beck is a psychiatrist at the University of Pennsylvania who really founded this whole line of work. And according to Beck, this — now, this applies to depression alone. It doesn't really apply to bipolar disorder. People who are depressed have a negative view of the self, of the future and of the world, and he calls this the "negative cognitive triad." And this negative cognitive triad is fed by specific cognitive deficits or biases. So, depressed people show a lot of distortions in thinking. "All-or-nothing thinking" is thinking that things are either all good or all bad. They can't sort of see the gray areas in between. There's "emotional reasoning," and an example of that is just if I feel like a loser I must be a loser. Of if I feel stupid, I must be stupid. And then "personalization," that is the self-blame that you often see in depressed people. And these kinds of distortions in thinking, these distorted ways of interpreting situations, feed a kind of general negative view of the self and hopelessness about the future.

People who are depressed also make attributions for negative events that are internal, that is, they blame themselves — that are stable. They see bad things as lasting forever and that are global. They see bad events as affecting many areas of their life, which, again, feeds their depression and their general assumption that life is terrible. And evidence that these — for these cognitive theories, that these negative cognitive styles predict depression — one of the best studies predicting this was a study that was jointly done at Temple University and the University of Wisconsin where they identified first year college students with a negative cognitive or attributional style. But these were people who had never experienced an episode of depression yet. They then tracked them for the next two years, and the bars here — the red bars are the percent of those with a negative cognitive style who developed an episode of major depression in that two and a half years versus the percent of those without a negative

cognitive style. And as you can see, there's quite a substantial difference between the two. So prospectively, these characteristics seem to predict your risk for depression.

In turn, there is a cognitive behavioral therapy that's based on Beck's theory. And the major steps in this involve identifying the themes in a person's negative thoughts and triggers for them and in helping the person challenge those thoughts by asking them what the evidence is for their interpretations, whether there are other ways of looking at the situation, how they could cope with the situation if a bad thing really did happen. So, the therapist helps the client recognize negative beliefs or assumptions and then challenges the truth value of these, and then change aspects of the environment that are related to depressive symptoms. So, they challenge your rational thinking, but they also recognize that there are really bad things that sometimes are going on the life of a person who is depressed, and they help them engage in more active problem solving to change those environments. They also teach the person ways to manage their mood so that they don't tumble down into depression. And these cognitive behavioral therapies have been shown to be extremely effective and as effective in some ways as the drug treatments.

So, this is a recent study in which they had 240 patients with major depressive disorder. They gave them four months of acute treatment with either cognitive behavioral therapy or Paxil, which is an SSRI. And in eight weeks here — they also had a placebo control group where they just got a pill, but it was a sugar pill. At eight weeks, the Paxil group, which is in red, and the CBT group, which is in yellow, were relatively even, although the Paxil group had a little bit of an edge over the CBT group. But by sixteen weeks the Paxil and CBT group were absolutely even in terms of the percent of people who were no longer depressed. So, both of them resulted in about 60% of people not being depressed.

And then one of the things that's been found over and over again with cognitive behavioral therapy is that it not only helps people get out of a current episode of depression, it helps to prevent future episodes of depression because it teaches the person new coping skills for dealing with new stressors that arise. So, in this same study they followed the patients for an additional twelve months. And of the Paxil group, half were left on full-dose medications to see if you could prevent depression by just keeping people on the meds, and half of the Paxil group were withdrawn to a placebo. And let's look at the placebo group first here.

This is the sad news about depression. If you just take drugs to get out of a current episode and you go off of the drugs without having any kind of psychotherapy, your rate of relapse tends to be very high. So, in this group of 240, almost 80% relapsed in the first year after going off of the active medication. Again, these were people who did not get any cognitive behavioral therapy. Of those people who stayed on Paxil alone, about 50% relapsed. But of those people who got cognitive behavioral therapy, only about 35% relapsed. And this kind of finding has been replicated over and over again, namely that CBT can reduce the rate of relapse in depression quite dramatically.

I want to give you a flavor for what CBT looks like, and I've got a short clip of a therapist who is actually, interestingly, the daughter of Aaron Beck and his heir in terms of the practice and development of cognitive behavioral therapy. And she's demonstrating CBT. This is actually a role play, but it's a pretty realistic role play, of how she goes after and helps to challenge a gentleman's negative cognitions about himself. This is a guy in the role who's recently lost his job and is really depressed over his job loss. [video clip playing]

Okay. I'm going to stop there because we're running out of time. But I just want to comment on a couple of things that she's doing. So, you see that she's having him generate his own challenges to his negative thoughts and write them down. And the whole — one of the major premises in CBT is it's not just what goes on in the therapy session that is effective. In fact, that's a minor part of it, but what the person practices in the time between therapy sessions. And so, what she's doing is helping him come up with a series of phrases he can say to himself when he feels discouraged and plunging down. She's also helping him do what's called "anticipatory coping," anticipating those situations that are going to trigger negative feelings and negative thoughts and coming up with ways of combating them in the moment that he can enact at the time.

Cognitive behavioral therapy is very structured, it's very focused and it's designed to be relatively short-term. The one other major kind of psychotherapy for depression is interpersonal therapy. It's based on the theory that negative views of the self and expectations about the self and relationships are based on upbringings that really fostered these kinds of negative self-views. And so, what you need to do is to help clients understand their negative self-views and how they're rooted in their past relationships. Interpersonal therapy is less structured than cognitive behavioral therapy, and it's more focused on the past. CBT is very focused on the present and dealing with the current situation that you're facing and combating that and developing coping styles for that.

There are a few studies comparing interpersonal therapy with CBT, but much less research has been done on IPT than CBT. But it is a positive alternative for some people, particularly those who find that their depression is very tied-up in recurrent themes in their relationships that seem to happen over and over and over again.

But to end, the good news about depression is that there are these alternative therapies. There is a number of drug therapies and there are at least two psychotherapies that good controlled research has shown can be very helpful. So, people have a choice, and there is absolutely no need for people to stay in depression but rather to seek out the kind of therapy that applies to and appeals to them the most. Good. Thanks very much.